Cloverb: Clock-Aware, Long-History Human Move Prediction with an Efficient Square-Token Transformer

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Abstract

Maia-3 (Chessformer) is the state of the art in predicting human chess moves. We reproduce it exactly, benchmark all nine released checkpoints, and audit the paper and code: we find that its public engine runs without the move history it was trained with (costing 2.2 points of accuracy), that its history encoding and Elo conditioning waste most of the input, that it ignores the clock, and that its headline ablation omits the baseline needed to credit its dynamic attention bias. We then present **Cloverb**, a square-token transformer conditioned on exactly what a player has at the board: their own rating, the time control, the time left on their clock, and the last 64 plies encoded as moves. A simple prior, initialising static attention biases with chess movement patterns, makes Cloverb learn 5.6 points faster than the Maia-3 architecture at equal data and makes Chessformer's dynamic bias generator unnecessary. With a C data pipeline, a fused AVX-512 attention kernel and bf16 training on Intel AMX, all experiments, including the final model distilled from Maia-3 through amortised teacher pools, ran on a single 4-core CPU.

1 Introduction

Predicting the move a human will play, rather than the best move, underpins human-like engines, tutoring and analysis tools. The Maia line of work [1, 2] and ALLIE [3] established the task; Chessformer [4] (released as Maia-3) is the current open state of the art at 57.1% move-matching accuracy with 79M parameters.

We started by reproducing Maia-3: we downloaded all nine released checkpoints, rebuilt the exact 884,049-position evaluation protocol, and benchmarked every variant (top-1, top-3, cross-entropy). The numbers reproduce, but reading the paper next to the released code exposed a series of inaccuracies and design decisions that leave accuracy and efficiency on the table (Section 3). Three matter most for a human-move model: (i) the network never sees the *clock*, and the protocol discards every position played under time pressure; (ii) history is eight board snapshots that alternate orientation every ply and are fused by a single linear map, which is why "more history does not help"; (iii) skill is encoded by concatenating two 128-d interpolated vectors to every one of the 64 tokens, i.e. two scalars occupying 73% of the input.

Cloverb keeps what works in Chessformer (squares as tokens, a from–to policy head) and re-designs the rest around the information a human player actually has and around efficiency:

- **Inputs a player sees.** The player's own Elo (not the opponent's), the time control (base seconds and increment), the time left on the player's clock, and the last ≤ 64 plies as *moves* scatter-added onto the squares they touched, all in one mover-relative frame; castling rights live on the rook squares and the en-passant square is marked.
- **Conditioning where it acts.** Elo and clock features form one vector that modulates every block through adaptive normalisation (adaLN-Zero [7]) instead of being copied 64 times.
- **Geometry as a prior, not a generator.** Every head of every layer owns a static 64×64 attention bias, *initialised* with chess movement patterns (rook lines, diagonals, knight jumps, king steps, pawn captures). This one change makes Cloverb learn 5.6 points faster than the Maia-3 architecture at equal data; Chessformer's dynamic bias generator turns out to be unnecessary.

Table 1: Released Maia-3 checkpoints on 100,016 positions of the ALLIE test set under Maia's protocol (first 10 plies and time pressure removed). "Paper" is the accuracy reported on the full 884,049 positions. 95% binomial half-width ≈0.31 points.

Checkpoint	Params	Paper top-1	Top-1	Top-3	CE (nats)
Maia-3 79M	78.9M	57.1	56.82	83.51	1.302
Maia-3 23M-ponder (clock)	23.0M	–	56.53	83.32	1.311
Maia-3 23M	22.9M	56.6	56.32	83.09	1.321
Maia-3 5M	5.2M	55.4	55.27	82.33	1.365
Maia-3 3M (ablation)	3.2M	54.8	54.58	81.77	1.389
5M, absolute PE	4.6M	54.7	54.37	81.62	1.395
5M, relative bias	4.6M	54.6	54.47	81.52	1.396
5M, no history	5.2M	54.0	53.79	81.34	1.410
5M, 32 positions	5.3M	55.4	55.15	82.26	1.369
5M, UCI default (history off)	5.2M	–	53.04	80.80	1.437

- **Systems.** A C chess core that parses Lichess PGN at >0.3M plies/s and builds training batches by replaying games on the fly (3.6 μs/sample, 6 bytes of storage per ply); a fused AVX-512 attention kernel for 64-token sequences; bf16 training on Intel AMX with `torch.compile`; jemalloc. Everything in this paper ran on *one 4-core CPU*.

2 Background

Maia [1] trained one CNN per rating band; **Maia-2** [2] unified them with skill-aware attention over CNN channels; **ALLIE** [3] is a decoder-only transformer over move sequences with clock and Elo tokens; **Chessformer / Maia-3** [4] uses an encoder over 64 square tokens with Geometric Attention Bias (GAB), an attention bias generated from a compressed board representation mixing shared templates. **Allie 2.0** [5], a 5.6B-parameter mixture-of-experts trained on 75B tokens with eight GPUs, reports a 0.31-point top-1 gain over Maia-3-79M on Maia's protocol and shows that the clock matters most in time trouble.

3 Reproducing and auditing Maia-3

Protocol. The ALLIE test set [3] contains 18,239 Lichess blitz games from 2022. Dropping each game's first 10 plies and every ply from the first one whose mover has fewer than 30 s selects *exactly* the 884,049 positions reported in the paper. We score a fixed sample of 100,016 positions (2,101 whole games) of that protocol with the official model code (bf16 trunk, fp32 policy head; Table 1). Our sample is about 0.25 points harder than the full set for every variant; all orderings and gaps of the paper's ablations are reproduced.

Findings. The full list with evidence is in the repository (`docs/critique.md`); the most consequential are:

1. *The released engine runs without history by default.* `maia3-uci` fills all history slots with the current position unless `--use-uci-history` is passed; this costs 2.2 points of top-1 accuracy, more than training without history at all.
2. *History is alternating and input-only.* Each past board is mirrored to its own side to move, so consecutive frames are flipped relative to each other, and all frames are fused by one linear map. The paper's conclusion that 32 positions are no better than 8 is a property of this encoding (we reproduce it: HIST32 vs. 5M), not of chess.
3. *No clock, and time pressure is removed from training and evaluation.* The unreleased-in-code MAIA3-23M-PONDER checkpoint, which does read the clock, is better even on the pressure-free protocol (Table 1).

4. *Elo as concatenated soft embeddings* is affine in the rating (the paper concedes it is equivalent to a scalar) yet fills 256 of 352 input channels on all 64 tokens.
5. *Missing state*: castling rights, en passant and the 50-move counter are absent from the human models although the paper lists them as necessary.
6. *Accounting*: "FLOPs" are multiply–accumulates (2× off); parameter counts do not match the checkpoints; an undocumented third ("ponder") head exists and one released checkpoint cannot be loaded by the released code; the value head is d wide, not 128; "all code and data" is open-sourced but no training code or data is.
7. *Missing baseline*: GAB is compared to absolute embeddings and to a translation-invariant 15×15 relative bias, but not to a static, per-square 64×64 bias, so the benefit of GAB being *dynamic* is not isolated (the paper's own Table 3 reports 0.77 cross-position correlation of GAB maps).
8. *Evaluation precision*: on CPUs, bf16 policy logits tie often enough that a strict rank count inflates top-1 by about one point (55.3%→56.4% for Maia-3-5M); we compute all policy heads in fp32.

4 Cloverb

4.1 Inputs

All squares are expressed from the mover's point of view (vertical flip and colour swap when Black moves). Token s starts as

$$x_s = E_{\text{piece}}[p_s] + E_{\text{sq}}[s] + \textstyle\sum_r \mathbf{1}[\text{right } r]\,\mathbf{1}[s = \text{rook}(r)]\,E_{\text{castle}}[r] + \mathbf{1}[s = \text{ep}]\,E_{\text{ep}} + \sum_{j=1}^{64} \big(\mathbf{1}[s = f_j]F_j + \mathbf{1}[s = t_j]T_j\big),$$

where (f_j, t_j) are the from/to squares of the move played j plies ago and $F_j = E_{\text{from}}[j] + E^{\text{from}}_{\text{pc}}[\text{side}_j, \text{piece}_j]$, $T_j = E_{\text{to}}[j] + E^{\text{to}}_{\text{pc}}[\text{side}_j, \text{piece}_j] + \mathbf{1}[\text{capture}_j]E_{\text{cap}}[\text{side}_j]$. The history term is a scatter-add of 128 vectors per position: it costs $O(64d)$ instead of a $(12n{+}256) \times d$ projection per token, it is lossless about the move sequence, and it lets the *same* square token accumulate everything that happened on it.

The conditioning vector $c = \text{MLP}(\phi)$ is built from: a hat-function (linear spline) encoding of the player's Elo on 15 anchors (400–3200), a hat encoding of $\log(1 + \text{time left})$ on 14 anchors (2 s–1 h), log base, log increment, time left relative to base + 40 inc, the halfmove and fullmove counters, castling rights, check, and the number of available history plies. c is added to every token and drives adaLN-Zero modulation (scale, shift and residual gate for attention and MLP) in every block.

4.2 Trunk and heads

Each block applies attention with RMS-normalised queries and keys [8] and an additive bias $B_{\ell h} = S_{\ell h} + \sum_{t=1}^{T} a_{\ell h t}(\bar{x}, c)\,U_t$: $S_{\ell h} \in \mathbb{R}^{64\times64}$ is a static learned bias per layer and head, and the optional second term mixes T templates shared by all layers with coefficients from the mean token and c (a cheap GAB analogue). The static biases are initialised with chess geometry: head 1 gets +2 on rook lines, head 2 on diagonals, head 3 on knight jumps, head 4 on king steps, head 5 on pawn-capture squares, head 6 on the same file, head 7 a $-0.5\cdot$Chebyshev-distance locality term, head 8 starts at zero (Section 5.2 shows that this prior is what matters and that $T{=}0$ suffices). The MLP is $W_2\,\text{ReLU}(W_1 h)^2$ [9]. Blocks are post-norm, as in Chessformer: $x \leftarrow \text{RMSNorm}(x + (1 + g_1)\,\text{Attn}(x(1 + s_1) + b_1))$ and likewise for the MLP, with (s, b, g) from adaLN-Zero; output projections are zero-initialised. The move history also records which piece was captured.

The policy head scores move $(u \to v)$ as $\langle W_f x_u, W_t x_v\rangle/\sqrt{d}$ over Maia-3's 4352-move vocabulary (64×64 plus 256 promotions, whose piece bias comes from the destination token), so Maia-3 checkpoints can serve as teachers. Two auxiliary heads read $[\bar{x}, c]$: game outcome (WDL) and think time (32 log-spaced buckets), the latter giving the engine a human-like time-management signal. The training loss is $\mathcal{L} = \text{CE}_{\text{legal}}(\text{move}) + 0.1\,\text{CE}(\text{WDL}) + 0.05\,\text{CE}(\text{time})$, with the softmax restricted to legal moves.

Table 2: Equal-data comparison on the ALLIE 10k sample (EMA weights): top-1 (%) / cross-entropy after N training positions, same data stream and evaluation for all runs; throughput is end-to-end training positions per second on the 4-core CPU.

Run	pos/s	0.51M	1.02M	1.54M
Chessformer-5M (Maia-3 architecture)	300	38.8 / 2.09	43.6 / 1.85	45.9 / 1.74
Cloverb d256 L8 (no teacher)	315	37.5 / 2.14	41.7 / 1.94	44.7 / 1.82
Cloverb + online Maia-3-23M teacher	153	42.9 / 1.90	–	–

4.3 Systems

Data. Games are stored whole: per ply a 16-bit move, the 16-bit clock before the move and the 16-bit time spent; per game ratings, time control and result. A C core parses Lichess SAN movetext with clock comments (validated move-for-move and clock-for-clock against `python-chess`), and builds batches by replaying each sampled game to the sampled ply, emitting the board, castling/en-passant state, the 64-ply history and a legal-move bitmask in $\approx 3.6\,\mu$s per sample on one core, in a background thread (ctypes releases the GIL). 5.5M games (372M plies) occupy 2.2 GB and any position can be sampled in any epoch.

Compute. Training uses bf16 autocast on Intel AMX with `torch.compile`. Profiling showed that GEMMs were only a third of a step: attention over $B \cdot H$ tiny 64×64 problems and memory-bound pointwise kernels dominated. PyTorch's fused CPU attention reached only ≈ 26 GFLOP/s on this shape, so we wrote a fused AVX-512 kernel that keeps each head's 64×64 scores in L1, fuses the static and template biases, and recomputes scores in the backward pass (custom `torch.library` op, exact to $2 \cdot 10^{-6}$ against PyTorch in fp32). Replacing glibc malloc by jemalloc removed page-fault storms on large tensors (+30% with our first kernel; once the projection layout and QK normalisation were fused into the kernel, the two allocators were within 1%). jemalloc must then run with a single arena: with per-thread arenas and decay disabled, retained memory under the compiled step grew by ≈ 0.3 GB per minute, and the first attempt at the final run was killed for lack of memory after 40 minutes.

5 Experiments

Data. Training uses four Lichess parquet shards [11] from August and September 2025 (after Maia-3's training window, January 2023–July 2025): 5.54M rated games with clocks from bullet to classical, 372M plies. A position is drawn by picking a (game, side) pair with weight $p(\text{Elo bin})^{-0.3} \times w_{\text{tc}}$ (bullet 0.6, blitz 1, rapid 1.3, classical 4) times the number of moves that side played, then a uniform ply of that side; plies before move 10 are kept with probability 0.5. Unlike Maia-3 we keep time-pressure positions.

Test sets. (1) The ALLIE 2022 blitz test set under Maia's protocol (100,016-position sample, Table 1); (2) the same 2,101 games with *all* 141,390 plies, i.e. including openings and time pressure; (3) fresh games from September 2026 (120k games streamed from the official dump, never seen by any model): 25k positions each of bullet, blitz, rapid and classical (all plies), and 50k blitz positions under Maia's protocol.

5.1 Equal-data comparison with Chessformer

We trained Cloverb (d=256, 8 layers, MLP×2, FLOP-matched to Maia-3-5M: 0.30 vs. 0.32 GMAC per position) and the authors' own Chessformer-5M module (inputs exactly as in `maia3.uci`) on the identical stream of 1.54M positions with the same optimiser, schedule shape and evaluation (Table 2). Two findings shaped the final system.

First, Chessformer's recipe is the bottleneck, not its architecture. With a sensible learning rate, preprocessing and kernels, the published 5M architecture learns fast: 45.9% after 1.5M positions, i.e. 83% of the way from chance to its published accuracy after 0.3% of its published training data. The first Cloverb configuration learned *more slowly per position* (44.7%) even though it trains 5% faster per second; Section 5.2 isolates why.

Table 3: Single-factor diagnostics after 512k training positions (EMA weights, ALLIE 10k sample; same data stream for every row). Indented rows each change one factor of the geometry-initialised model.

Configuration	Top-1	Top-3	CE
Chessformer-5M (authors' module)	38.8	63.4	2.088
Cloverb, zero-initialised biases	37.5	61.5	2.140
Cloverb + geometry initialisation	44.4	71.1	1.823
+ post-norm blocks	45.0	72.4	1.791
+ captured-piece type in history	44.7	71.5	1.818
without dynamic templates (T=0)	44.4	71.1	1.823

Second, a teacher is worth more than data. Distilling the released Maia-3-23M online (soft targets with weight 0.7 plus the human move) lifts Cloverb from 37.5% to 42.9% at 0.5M positions, beyond what the student reaches with twice the data. Because the teacher's forward pass costs about as much as a student training step, we amortise it: the final model trains for several epochs on a fixed pool of positions labelled once by the clock-aware Maia-3-23M-ponder checkpoint (top-16 moves, which hold 99.1% of the probability mass).

5.2 Diagnosing the per-sample gap: geometry is the missing prior

We ran single-factor diagnostics for 512k positions each on the identical data stream (Table 3). One change dominates everything else: *initialising the static attention biases with chess geometry.* Each of the first heads of every layer starts with a +2 logit bonus on the squares reachable by one movement pattern (rook lines, bishop diagonals, knight jumps, king steps, pawn-capture squares, same file) and one head with a Chebyshev-distance penalty; the biases remain fully trainable. This single prior lifts Cloverb from 37.5% to 44.4% after 512k positions, *5.6 points above the Maia-3 architecture* at equal data, and reaches what the zero-initialised model needed three times the data for. It also answers the paper's missing baseline: most of what GAB contributes early in training is geometry, which a static, per-square bias with the right initialisation supplies at a fraction of the cost. Post-norm blocks (as in Chessformer) add a further 0.6 points on top of the geometric prior at 6% lower throughput; encoding *which* piece was captured in the move history adds 0.3 points for free; and the dynamic template mix contributes nothing (44.35% without vs. 44.37% with, equal cross-entropy), so the final model drops it and keeps only static, geometry-initialised biases.

5.3 Final model

Recipe. The final model has 9.22M parameters: $d = 256$, 10 post-norm blocks with 8 heads, ReLU^2 MLPs of width $3d$, geometry-initialised static attention biases, no dynamic templates, and the captured piece in the move history. Six pools of 2M positions each, drawn with the training sampler, were labelled once by Maia-3-23M-ponder (top-16 moves; about 85 minutes per pool at 390 positions/s). Training uses AdamW ($\beta = (0.9, 0.98)$, weight decay 0.05, learning rate 10^{-3}, 1k warm-up steps), batch 512, gradient clipping at 1, an EMA of the weights with decay 0.9998, and bf16 autocast, at about 280 positions/s.

Switching to pure distillation mid-run. For the first 24k steps the target mixed the teacher's distribution (weight 0.7) with the human move (0.3); below 30 s on the mover's clock, where the teacher never trained, the mix was 0.4/0.6. Projecting the learning curve showed this would land near Maia-3-5M, so at step 24k we branched two 3k-step learning-rate cool-downs from the same checkpoint on identical batches. With the human move removed outside time trouble, the cooled-down model reached a cross-entropy of 1.3970 and 54.36% top-1, against 1.3996 and 54.19% with the mixed target. Its raw (non-EMA) cross-entropy was already 0.008 lower halfway through the cool-down, consistent with less gradient noise. The run continued from step 24k with pure teacher targets (still 0.4/0.6 under 30 s) to 80k steps, with the learning rate decaying linearly to 2% over the last 30%. In total the model sees about 41M samples, 3.4 passes over the pool.

Table 4: Maia-3 checkpoints and their ensembles as teachers, on the 10k-position evaluation sample (Maia protocol). Ensembles average the members' log-probabilities ("logits") or probabilities.

Model	Top-1	Top-3	CE
Maia-3 79M + 23M + 23M-ponder (logits)	56.94	84.22	1.288
Maia-3 79M + 23M-ponder (logits)	56.84	84.16	1.285
Maia-3 79M	56.66	84.36	1.290
Maia-3 23M	56.58	83.74	1.307
Maia-3 23M-ponder (our teacher)	56.49	83.86	1.299
Maia-3 5M	55.25	82.75	1.351

How far distillation can go. A student trained to match a teacher approaches the teacher's accuracy from below. Table 4 scores every Maia-3 checkpoint and their averages on the evaluation sample used during training. The clock-aware teacher we use is 0.2 points below Maia-3-79M; the best ensemble (79M, 23M and 23M-ponder, averaged logits) is 0.3 points above it. Distillation alone can therefore at most edge past the strongest Maia-3 under Maia's protocol. Beating it clearly needs what a teacher cannot supply: far more human moves, or information the teacher does not see. Cloverb's clock, increment and 64-ply history are such information, and they matter most in exactly the positions Maia's protocol removes.

5.4 Inference

Cloverb ships as a single C file with no Python or PyTorch dependency: the C chess core encodes the position and its history, the trunk's GEMMs run on AMX bf16 tiles with weights packed once into VNNI tiles, attention runs in AVX-512 with each head's scores kept in registers and L1, and the policy head stays in fp32. Against the PyTorch forward pass of the same weights it reproduces move probabilities to within 10^{-3} and the same top move on every tested position.

6 Limitations

- **Compute.** Every number here comes from one 4-core CPU over a few days; Cloverb saw tens of millions of positions, Maia-3 half a billion. Our conclusions about architecture are about learning speed in this regime; at Maia-3's budget the geometric prior's advantage may shrink.
- **Teacher.** The final model is distilled from a released Maia-3 checkpoint (with the human move as a second target), so its accuracy partly inherits Maia-3's; the equal-data comparisons in Tables 2 and 3 use no teacher. Maia-3 never trained on positions with less than 30 s on the clock, so in time trouble the human move keeps a weight of 0.6 in our loss (elsewhere 0.3 for the first 24k steps and 0 afterwards).
- **Opponent rating.** By design Cloverb sees only the mover's rating, as an engine user would set it; Maia-3 also sees the opponent's.
- **Test sets.** The ALLIE test set is blitz from 2022; our training games are from 2025. The fresh 2026 test set covers all time controls but has few classical games (615).

7 Conclusion

Reproducing Maia-3 confirmed its accuracy and exposed avoidable losses in its inputs, defaults and evaluation. Cloverb reorganises the same square-token idea around what a human sees at the board (own rating, clock, the moves that led here) and around the hardware it runs on, and shows that a simple geometric prior on attention, not a dynamic bias generator, is what lets such a model learn chess quickly.

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